

FEB12001X – Exam 25 October 2024

FEB12001X Applied Microeconomics 24/25 (ESE) · 9 exercises · 26.0 points

1 Question 1

4.0 points · 3 questions

Consider a not-for-profit organization that aims to maximize the amount of social goods it provides. It can buy social goods in the market at price p per unit. It can also hire workers to produce social goods. Each worker produces an amount of Q social goods. The number of workers willing to work at the organization depends on the wage w offered by the organization and is described by the following function: $N(w)$. The first derivative of this function is strictly positive. Moreover, it holds that $N(0) > 0$. The organization must pay all workers that it employs the same wage (in other words, wage discrimination is not possible). The organization has a budget equal to B that it fully spends.

Text

a Provide an expression for the budget constraint of this organization.

1.0 point · Open · 3/10 Page

+1 point

$$B = wN(w) + pZ$$

b Describe in at most two clear sentences what it means that $N(0) > 0$ (provide a clear economic interpretation, do not provide "math in words").

1.0 point · Open · 2/5 Page

+1 point

It means that when the organization offers a wage equal to zero, some people want to work for the organization. You can also say: some people wish to volunteer for the organization.

Assume that circumstances are such that it is in the interest of the organization to buy some social goods in the market and produce some social goods by hiring workers.

Text

c Derive the first-order condition describing the level of the wage the organization should offer so as to maximize the amount of social goods it provides. Provide an economic interpretation for each of the terms in the first-order condition (as usual: write clear sentences, and no math in words).

2.0 points · Open · 3/4 Page

+1 point

The amount of social goods is $N(w)Q + Z$. Use the budget constraint $B = wN(w) + pZ$ to replace Z . Differentiating to w and setting equal to zero gives the first-order condition: $N'(w)Q - \frac{1}{p}[N(w) + wN'(w)] = 0$.

+1 point

The first term says that when the wage goes up, more people can be hired, each produces Q , and so production goes up. The second term says that as the wage is increased, the total wage bill increases, leaving less of the budget to buy social goods in the market.

2 Question 2

5.0 points · 5 questions

Arno and Bea jointly own a company. The revenues of the company R depend on how much effort they exert. This is described by the following function: $R = p(e_a + e_b)$, where e_a is Arno's level of effort and e_b is Bea's level of effort. Arno and Bea have decided to share the revenues of the company equally. Arno and Bea have no other income than their share in the revenues of the company. Arno's utility equals $U_a = Y_a - \frac{1}{2}\theta e_a^2$ and Bea's utility equals $U_b = Y_b - \frac{1}{2}\lambda e_b^2$. The first term in each utility function reflects the utility from income. The second represents the costs of effort. Assume that $\theta > \lambda$. Arno and Bea choose their effort independently and each of them aims to maximize his/her own utility.

Text

a What level of effort does Arno choose?

1.0 point · Open · 1/2 Page

+1 point

Arno maximizes $U_a = \frac{1}{2}p(e_a + e_b) - \frac{1}{2}\theta e_a^2$. Differentiating to e_a and setting equal to zero gives: $\frac{1}{2}p - \theta e_a = 0$. Solving gives $e_a = \frac{1}{2}\frac{p}{\theta}$.

b Define social welfare as the sum of utilities of Arno and Bea. Which levels of effort of Arno and Bea maximize social welfare?

1.0 point · Open · 1/2 Page

+1 point

Summing up the utility functions of Arno and Bea gives the following expression for social welfare: $p(e_a + e_b) - \frac{1}{2}\theta e_a^2 - \frac{1}{2}\lambda e_b^2$. Differentiating to e_a and to e_b respectively and setting equal to zero gives the first-order conditions: $p - \theta e_a = 0$ and $p - \lambda e_b = 0$. Solving gives $e_a = \frac{p}{\theta}$ and $e_b = \frac{p}{\lambda}$.

Arno and Bea think about changing the way they divide the revenues of the company. An economist proposes the following sharing rule: Arno gets share $\frac{e_a}{e_a + e_b}$ of revenues and Bea gets $\frac{e_b}{e_a + e_b}$. Assume that Arno and Bea can commit to this division of revenues (that is, we assume that there will be no conflict about how to divide revenues after they have exerted effort).

Text

c Derive the level of effort that Arno will choose under the sharing rule proposed by the economist.

1.0 point · Open · 3/5 Page

+1 point

Arno's utility becomes: $\frac{e_a}{e_a + e_b}p(e_a + e_b) - \frac{1}{2}\theta e_a^2$. The first-order condition is $p - \theta e_a = 0$ and so $e_a = \frac{p}{\theta}$.

The economist claims that the sharing rule that she proposes maximizes social welfare.

Text

d How can you check whether the claim by the economist is correct? Note: you do not need to check, you only need to describe what step(s) you would take to check the claim.

1.0 point · Open · 1/2 Page

+1 point

Derive the level of efforts that Arno and Bea choose under the new sharing rule and compare these levels to the levels of effort by Arno and Bea that maximize social welfare. If they are the same, we can conclude that the new sharing rule maximizes social welfare.

Another economist says that, even in the case that the proposed sharing rule maximizes social welfare, it is not sure that both Arno and Bea will agree with it. This economist says that Arno may be worse off under the proposed sharing rule as compared to the rule where each of them get half of the revenues because $\theta > \lambda$.

Text

e What analysis would you do to check whether Arno is worse off under the proposed sharing rule? Note: you do not need to check, you only need to describe what step(s) you would take to check the claim.

1.0 point · Open · 1/2 Page

+1 point

Derive what level of effort Arno chooses under the new sharing rule (that is, your result under 2c) and substitute this into his utility function. Derive also what level of effort Bea chooses under the new sharing rule and substitute this into Arno's utility function as well. Similarly, derive what level of effort Arno chooses under the equal division rule (that is, your result under 2a) and substitute this into his utility function. Also, derive the level of effort Bea chooses under the equal division rule and substitute this into Arno's utility function. Lastly, compare the two resulting utilities to see whether (or under what condition) Arno is better off.

3 Question 3

3.0 points · 2 questions

A profit-maximizing company hires a worker. The worker exerts effort e which results in output Q following the production function: $Q = \kappa e$. The company sells output at price P per unit, which is a constant. The worker's utility function is: $U = a + bQ - \frac{1}{2}\theta e^2 + \gamma(PQ - a - bQ)$, where U is the worker's utility level, a is the base salary paid to the worker, b is a bonus per unit of output, θ is a cost-of-effort parameter, and γ measures how much the worker cares about the company's profits. The worker chooses effort so as to maximize her utility.

Text

a If the worker works at this company, what level of effort does the worker choose? Derive the first-order condition and solve for worker's effort.

1.0 point · Open · 3/5 Page

+1 point

Substitute the production function into the utility function, differentiate to effort and set equal to zero: $b\kappa - \theta e + \gamma(P - b)\kappa = 0$. Solving gives $e = \frac{b\kappa + \gamma(P - b)\kappa}{\theta}$.

The model in this Question predicts that workers who care more about the profits of the company (that is, workers who have a higher γ) are less responsive to an increase in the bonus b (that is, they increase effort to a smaller extent when the bonus goes up).

Text

b Suppose you want to test this prediction in a large company with many production plants. For convenience, assume that plants are completely independent from each other and that productivity can be measured at the worker level. Each plant employs a large number of workers who differ in γ . How would you design your empirical study to test the prediction? Maximize the credibility of your study and ignore any feasibility constraints.

2.0 points · Open · 1/2 Page

+1 point

Run a randomized control trial where in randomly assigned production plants (the treatment group) the bonus b is increased whereas in the others (the control group) it remains the same.

[Optional (that is, what follows is not necessary to get the point): Before the trial starts, measure the γ of all workers in the company, e.g. by running a questionnaire or, even better, running an incentivized study where people can allocate money to the company at varying costs (the dictator game).]

+1 point

Compare workers' productivity in treatment and control plants to estimate the effect of the increase in the bonus. Study whether there is a smaller treatment effect among workers with higher γ .

4 Question 4

1.0 point · 1 question

A principal wishes to hire one agent for two periods. There exist two types of agents: skilled and unskilled. The principal wishes to hire a skilled agent, but she cannot observe an agent's type. Agents themselves do know their type. Skilled agents earn wage W_s per period when they work somewhere else. Unskilled agents earn wage W_u per period when they work somewhere else. All that agents care about is their earnings. They do not discount the future (i.e. the two periods have equal weight). At the end of the first period, the principal may discover the type of agent she hired. This happens with probability p . With the remaining probability $(1 - p)$, the principal does not learn the agent's type in period 1. We assume that if the principal discovers that the agent she hires is unskilled, she fires the agent. Such an agent works elsewhere and so earns W_u in the second period. In all other cases, the agent is retained. The principal pays wage W_1 in the first period (both when the agent is fired and when the agent is not fired). If an agent is retained, she earns wage W_2 in the second period.

Text

Under which two conditions on the wages W_1 and W_2 does the principal make sure that only the skilled agents wish to work for this principal?

1.0 point · Open · 1/2 Page

+1 point

The unskilled will not apply if: $W_1 + (1 - p) W_2 + p W_u < 2 W_u$. The skilled will apply if $W_1 + W_2 \geq 2 W_s$.

5 Question 5

3.0 points · 3 questions

Consider Anne, Ben, and Dikra, who all value their consumption of a pure public good as well as their consumption of a private good. The private good is traded at price 1, which is equal to marginal cost: $p = MC = 1$ euro. Everyone's marginal utility of the private good equals 1. Anne's marginal utility of the public good is $MU_A = \alpha$ (such that her $MRS_A = \frac{\alpha}{1}$). Ben's marginal utility of the public good is given by $MU_B = b$. Dikra's marginal utility of the public good is given by $MU_D = d$. Only Anne can provide units of the public good. Her total cost of providing Q units of the public good are given by $TC_A = \frac{1}{2}\lambda Q^2$. Assume that all three individuals maximise their own utility and that $\alpha > 0$, $b > 0$, $d > 0$, and $\lambda > 0$.

Text

a Determine the socially efficient level of provision of the public good.

1.0 point · Open · 2/5 Page

+1 point

Substituting the marginal utilities and marginal cost into $MRS_A + MRS_B + MRS_D = MRT$ yields $\frac{\alpha}{1} + \frac{b}{1} + \frac{d}{1} = \frac{\lambda Q}{1}$ which can be solved for $Q^* = \frac{\alpha+b+d}{\lambda}$. Only using the marginal utilities and cost of the public good ($\alpha + b + d = \lambda Q$) is also correct.

Suppose Anne decides independently how many units of the public good to provide. Then, the efficiency loss due to market failure is given by $\frac{1}{2\lambda}(b + d)^2$.

Text

b Explain why the efficiency loss does not depend on α .

1.0 point · Open · 7/20 Page

+1 point

Anne chooses a level of Q that maximises her individual utility. Hence, she does take her own marginal utility of public goods consumption α into account, but ignores the utility derived by Ben and Dikra.

c Suppose Anne and Dikra negotiate in an efficient manner about (i) the level of the public good provided by Anne and (ii) a monetary transfer from Dikra to Anne. Ben refuses to be part of these negotiations. Derive how much Dikra is maximally willing to pay to Anne.

1.0 point · Open · 17/20 Page

+1 point

These negotiations will yield the level of public goods provision that is efficient for Anne and Dikra together: $MRS_A + MRS_D = MRT$ yields $\frac{\alpha}{1} + \frac{d}{1} = \frac{\lambda Q}{1}$ which can be solved for $Q^N = \frac{\alpha+d}{\lambda}$. Absent negotiations, the outcome would be $Q^P = \frac{\alpha}{\lambda}$. Hence, Dikra is maximally willing to pay $d(\frac{\alpha+d}{\lambda} - \frac{\alpha}{\lambda}) = \frac{d^2}{\lambda}$.

0 points

Wrong way of calculating Q^N or Q^P or wrongly assuming that Dikra pays for all Q^N units instead of only the increase from Q^P to Q^N

6 Question 6

4.0 points · 4 questions

Consider the market for good X . Demand for good X is given by $P = \beta - 2Q_D$, where P is the price per unit of good X in euro, Q_D is the quantity demanded, and β is a parameter. Supply of good X is given by $P = Q_S$, where Q_S is quantity supplied. Production of good X yields external cost. Marginal external cost are constant and given by $c = 15$ euro. Assume that $\beta > 15$.

The government considers to ban the production of good X (such that $Q_S = Q_D = 0$).

Text

a Derive the condition (which depends on β) under which a ban on the production of good X reduces efficiency, compared to the situation without any government intervention.

1.0 point · Open · 3/4 Page

+1 point

With a ban, total value created by the market equals 0. Without a ban, the equilibrium is given by $\beta - 2Q = Q$ which yields $Q = \frac{1}{3}\beta$. Total value created equals $\frac{1}{2}\beta * \frac{1}{3}\beta - c\frac{1}{3}\beta = \frac{1}{6}\beta^2 - 5\beta = \beta(\frac{1}{6}\beta - 5)$. Hence, the ban reduces efficiency if $\frac{1}{6}\beta - 5 > 0$ which yields $\beta > 30$.

Equally correct: calculating consumer surplus, producer surplus, and external cost separately: In equilibrium, $Q = \frac{1}{3}\beta$ and $p = \frac{1}{3}\beta$. CS: $\frac{1}{2}(\beta - \frac{1}{3}\beta)\frac{1}{3}\beta = \frac{1}{9}\beta^2$. PS: $\frac{1}{2}(\frac{1}{3}\beta)\frac{1}{3}\beta = \frac{1}{18}\beta^2$. TEC: $15 * \frac{1}{3}\beta = 5\beta$. CS+PS-TEC yields the same outcome as above.

b Explain why this condition depends on β . Give an economic explanation, not math in words.

1.0 point · Open · 2/5 Page

+1 point

This market creates consumer surplus, producer surplus, and external cost. A higher value of β implies a higher consumer surplus. This makes a ban more costly in terms of efficiency.

0 points

Any answer that only argues that a higher β yields higher value because it implies a higher equilibrium quantity.

c Assume that $\beta = 150$. Suppose the government does not ban the production of good X , but introduces a unit tax of 30 euro per unit on the production of good X . Determine the share of the tax burden that is borne by consumers of good X .

1.0 point · Open · 1/2 Page

+1 point

Before-tax equilibrium: $Q = 50$ and $P = 50$. After-tax: $Q = 40$, price consumers $P_C = 70$, and price producers $P_S = 40$. Hence, consumers bear share $2/3$ of the tax burden.

d Explain why consumers bear a share of the tax burden even though the government imposes a tax on producers. Note: it is not sufficient to only mention the appropriate term; you have to provide an explanation.

1.0 point · Open · 2/5 Page

+1 point

The tax would drive down the price producers keep. However, producers have alternatives to producing and selling good X as embodied in their supply function / supply elasticity. By reducing quantity supplied, they can shift part of the tax burden to consumers (as a reduction in quantity supplied implies that marginal willingness to pay of consumers increases).

7 Question 7

4.0 points · 3 questions

A small city has 50 identical poor citizens and 50 identical rich citizens. On a plot of land in the city, a park can be constructed. The cost of the park are 20.000 euro. Each rich inhabitant values the park at 300 euro, and each poor inhabitant values the park at 160 euro.

The cost of constructing the park will be covered by levying a tax on champagne. Only the rich drink champagne. Demand for champagne by all rich citizens together is given by $P = 250 - \frac{1}{4}Q_R$, where P is the price of champagne and Q_R is the quantity demanded. The marginal cost of production of champagne are constant. In the absence of a tax, supply of champagne is given by $P = MC = 100$. The government levies a unit tax of 50 euro on champagne.

Text

a Derive the Excess Burden of this tax.

1.0 point · Open · 1/2 Page

+1 point

The tax raises consumer price to 150. Equilibrium without tax: $Q_C = 600$. Equilibrium with tax: $Q = 400$. Excess burden equals $\frac{1}{2}50(600 - 400) = 5000$ euro.

Equally correct (but more cumbersome): calculate demand elasticity in before-tax equilibrium: $\epsilon_D = \frac{\Delta Q}{\Delta P} \frac{P}{Q} = \frac{1}{\frac{1}{4}} \frac{100}{600} = \frac{2}{3}$. Observe that a unit tax of 50 is identical to an ad valorem tax $t = 0.5$ in this exercise. Substitute this into the formula $EB = \frac{1}{2}\epsilon_D PQ t^2$ yields $EB = 5000$.

The government maximises an additive social welfare function: $SW = 50U_R + 50U_P$, where U_R is the utility of a rich person and U_P is the utility of a poor person. The marginal utility of income of a rich person equals 1. The marginal utility of income of a poor person equals α .

Text

b Determine for which levels of α constructing the park (and levying the tax on champagne) leads to an increase in social welfare.

2.0 points · Open · 7/10 Page

+1 point

Total benefits of the rich in euro: $50 \cdot 300 = 15.000$. Total cost for the rich are the construction cost (= total tax burden) plus the excess burden: $20000 + 5000 = 25000$. Total benefits of the poor: $50 \cdot 160 = 8000$.

+1 point

Hence, social welfare increases if $15000 - 25000 + 8000\alpha > 0$ which yields $\alpha > 1.25$. ($\alpha \geq 1.25$ is equally correct)

c Suppose the government finds a way to tax the citizens without inducing an excess burden and to redistribute among citizens without any loss in efficiency (in other words: the Second Fundamental Theorem of Welfare holds). Explain how this affects the government's decision to construct the park.

1.0 point · Open · 2/5 Page

+1 point

Now, we can fully separate efficiency and distribution. As a result, it becomes socially optimal to construct the park if and only if it increases efficiency, as any undesirable distributional effects can be redistributed.

Equally correct: as the government can now redistribute any gains of the park without loss of efficiency, it is possible to create a Pareto-improvement as total benefits ($50 \cdot 300 + 50 \cdot 160 = 23000$) are larger than total cost.

(As a sidenote: under these conditions, the government may engage in extensive redistribution from rich to poor, which in turn could affect their valuation of the park in monetary terms. If so, the first answer is still correct)

8 Question 8

2.0 points · 2 questions

Consider a city that has to decide on how much to spend on road maintenance. Let m denote the amount of spending, in million euro. All citizens have single-peaked preferences regarding the amount of spending m . The citizens can be grouped into three equally large groups. Group North prefers to spend $m_N^* = 20$. Group West prefer to spend $m_W^* = 40$. For group East, the utility of spending m million euro on road maintenance can be given by

$$U_E = 5 \ln(m) - \frac{1}{3}m$$

.

Text

a Derive the optimal amount of spending m_E^* for group East and show that the preferences for group East are single-peaked.

1.0 point · Open · 7/10 Page

+0.5 points

Maximising U_E with respect to m yields first-order condition $\frac{5}{m} - \frac{1}{3} = 0$ which yields $m_E^* = 15$.

+0.5 points

Taking the derivative of the first-order condition with respect to m yields second order condition $-\frac{5}{m^2} < 0$. Hence, utility U_E has an inverted U-shape with one maximum: the single peak.

b Suppose the decision on m is made by pairwise majority voting. Which level of m wins against all other possible levels of m ? Explain your answer.

Note: if you could not derive the optimal amount of spending by group East m_E^* , start your answer here with assuming some level of m_E^* .

1.0 point · Open · 2/5 Page

+1 point

If everyone has single-peaked preferences and pairwise majority voting is used, then the option preferred by the median voter wins against all alternative. Given that $m_N^* = 20$, $m_W^* = 40$, and $m_E^* = 15$, group North is the median voter, such that $m = 20$ wins against all alternatives. Note: if you have a different level of m_E^* and used that correctly to determine which group is the median, your answer is correct here.

9 Additional answer space

0.0 points · 5 questions

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found**. Otherwise, the answers in the additional space below will not be reviewed.

NB: Do **not** use this additional space for answers that are too long.

Text

a Additional answer box 1

Open · 4/5 Page

b Additional answer box 2

Open · 1/2 Page

c Additional answer box 3

Open · 1/2 Page

d Additional answer box 4

Open · 1/2 Page

e Additional answer box 5

Open · 1/2 Page